USER & SERVICE MANUAL

DR Rene's Magic Toolbox

Version 2.0.2

Diagnostics, maintenance, recovery, and controller event-log collection

FOR TRAINED EMPLOYEES ONLY

Zwart Industrial Innovations

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Document information

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> **What's new in this release:** LGC EventLog Puller and NGC EventLog Puller have been added. Both modules collect controller event logs and save them locally in a password-protected ZIP archive. The main interface image in this manual has also been updated.

Contents

1. Introduction

The DR Rene's Magic Toolbox (DRRMT) is a service and maintenance application for Doosan Robotics systems and related components, including controller generations, vision systems, and supporting subsystems. It provides a structured interface for diagnostics, maintenance, recovery, and service-data collection.

> **Authorized use:** This tool may only be used by trained employees and authorized service personnel.

2. Important Disclaimer

> **Use at your own risk:** Some modules can make permanent changes or permanently delete controller data. Read the information panel and warning text before starting any operation.

- Zwart Industrial Innovations is not liable for damage, data loss, downtime, or malfunction caused by use of this tool.
- The user is fully responsible for selecting the correct robot, controller type, target IP address, and module.
- Create a backup before any destructive or version-manipulation operation whenever a backup is possible.
- Do not interrupt the controller or workstation while an operation is running unless the module instructs you to do so.

3. Installation

1. **Run** the supplied MSI installer.
2. **Follow** the installation wizard and accept the applicable license terms.
3. **Launch** DR Rene's Magic Toolbox from the desktop shortcut or the Start menu.
4. **Confirm** that the application opens and that the version displayed in the upper-left corner is correct.

4. Main Interface Overview

The main interface contains the software version and release date, a shared Robot IP Address field, a status line, and the available service modules. The saved IP address is used by all modules that connect to a controller.

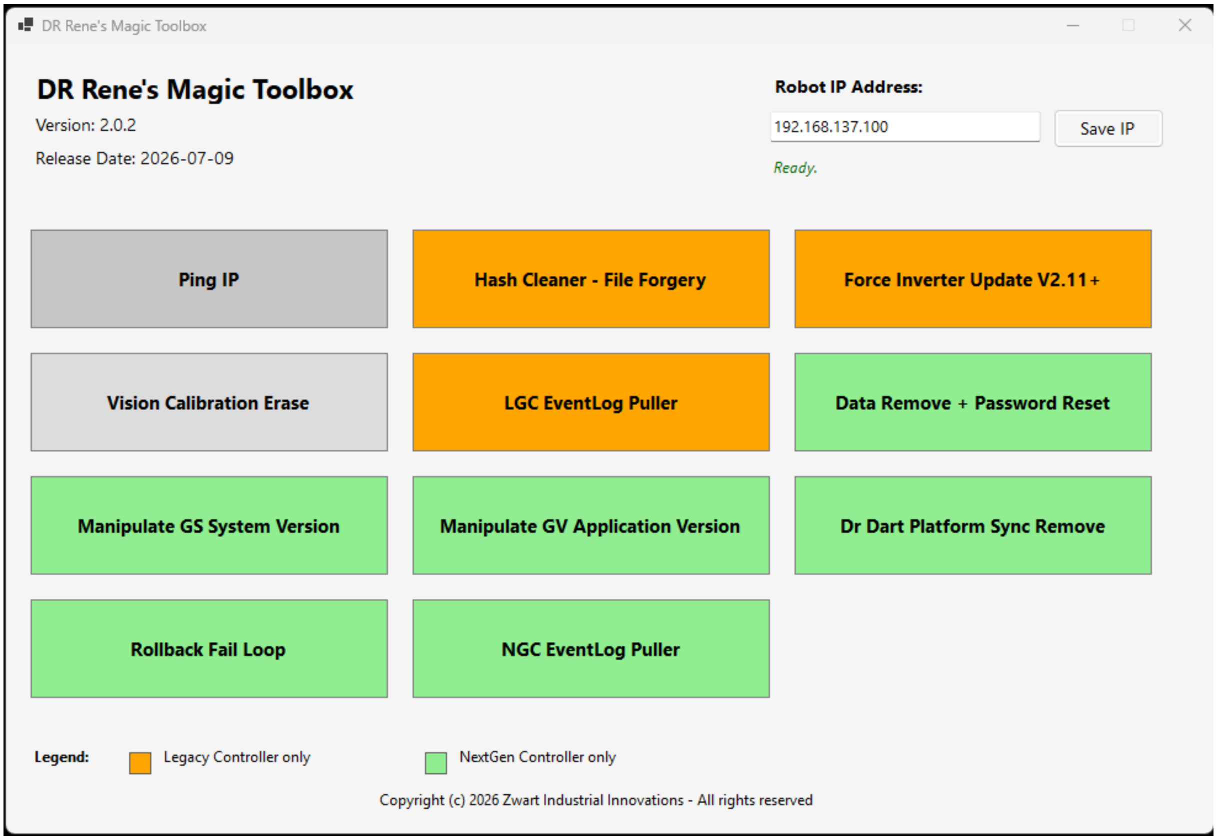

Figure 1. DR Rene's Magic Toolbox main interface (Version 2.0.2).

- **Software information:** shown in the upper-left corner.
- **Robot IP Address:** shared target address used by controller modules.
- **Status:** confirms whether the main interface is ready.
- **Module buttons:** open the selected service, maintenance, recovery, or log-collection function.

5. Setting the Robot IP Address

1. **Enter** the robot or controller address in the Robot IP Address field.
2. **Select** Save IP.
3. **Verify** that the status remains Ready and that the correct address appears when a module opens.

> **Validation:** The IP address is checked before it is saved. Invalid IP addresses are rejected.

6. Button Colors (Legend)

Color	Meaning
Orange	Legacy Controller (LGC) only
Green	NextGen Controller (NGC) only
Gray / default	General function

> **Controller selection:** Always choose the module color that matches the connected controller generation.

7. General Module Workflow

1. **Confirm** the saved Robot IP Address and controller generation.
2. **Select** the required module from the main interface.
3. **Read** the complete information panel and any warnings.
4. **Choose** any required destination folder or confirm the requested action.
5. **Run** the module and monitor the result panel until a final success or failure message appears.
6. **Perform** the indicated reboot, update, verification, or archive handoff when required.

8. Module Descriptions

8.1 Ping IP

Tests network connectivity to the saved Robot IP Address. The module performs four ping attempts and reports whether the target responded.

8.2 Hash Cleaner - File Forgery

Legacy Controller only. Use this module when the robot reports FILE FORGERY FOUND. Restart the controller after the function completes.

> **Service note:** If the issue remains after using Hash Cleaner, SSD corruption may be the underlying cause and further diagnosis is required.

> **Irreversible action:** Only run this module when the reported fault and controller type have been verified.

8.3 Force Inverter Update V2.11+

Legacy Controller only. Forces the inverter update state so the inverter firmware can be brought into line with the ControlBox and SafetyBoard versions.

Requirements:

- Inverter version 2.11 or later
- ControlBox version 2.11 or later
- SafetyBoard version 2.11 or later

Example: when the ControlBox and SafetyBoard are on V2.12.2 and the inverter is on V2.11.1, the module can prepare the inverter to update to the matching V2.12.2 data.

> **After execution:** Reboot the robot. During startup, the robot indicates that it is being prepared while the new inverter data is written to the PCBs.

8.4 Vision Calibration Erase

Connects to the vision camera and permanently removes its calibration data.

- Four ping checks
- SSH connection test
- Verification after cleanup

> **Permanent deletion:** Calibration data cannot be recovered through this module.

8.5 LGC EventLog Puller

Legacy Controller only. Collects matching event-log files from the controller without deleting or modifying the controller logs, then packages the downloaded copies in a local password-protected ZIP archive.

Property	LGC EventLog Puller
Main-menu color	Orange
Controller generation	Legacy Controller (LGC)
Output filename	EventLogs_YYYYMMDD_HHMMSS.zip
Archive	Password-protected ZIP saved in the selected customer folder

Procedure

1. **Save** the Legacy Controller IP address in the main window.
2. **Select** LGC EventLog Puller.
3. **Check** the Target IP shown in the module window.
4. **Select** Browse and choose an existing customer folder with write access.
5. **Select** Run EventPuller.
6. **Wait** until EventPuller completed successfully appears.
7. **Record** the full ZIP path shown in the result panel and transfer the archive according to the service procedure.

> **Archive handling:** The archive may contain customer and diagnostic data. Keep it protected and share it only through approved channels. Obtain the ZIP password through the authorized internal process.

8.6 Data Remove + Password Reset

NextGen Controller only. Resets the system to a default state, including a password reset and loss of configuration, programs, and other controller data.

> **Permanent deletion:** Back up required data before execution. This module is destructive.

8.7 Manipulate GS System Version

NextGen Controller only. Sets the reported GS system version to GS03020000_20240305_124 so the controller can be rebooted and updated to the required GS version.

> **Manual update required:** Reboot the controller and perform the intended software update after using this module.

8.8 Manipulate GV Application Version

NextGen Controller only. Sets the reported GV application version to GV03020000_20240315_211 so the controller can be rebooted and updated to the required GV version.

> **Manual update required:** Reboot the controller and perform the intended software update after using this module.

8.9 Dr Dart Platform Sync Remove

NextGen Controller only. Removes synchronization data. This is required after manipulating the GS or GV version data.

> **After execution:** Reboot the controller before continuing.

8.10 Rollback Fail Loop

NextGen Controller only. Repairs rollback-loop conditions that can occur after a failed software update.

8.11 NGC EventLog Puller

NextGen Controller only. Collects matching event-log files from the controller without deleting or modifying the controller logs, then packages the downloaded copies in a local password-protected ZIP archive.

Property	NGC EventLog Puller
Main-menu color	Green
Controller generation	NextGen Controller (NGC)
Output filename	EventLogs_NGC_YYYYMMDD_HHMMSS.zip
Archive	Password-protected ZIP saved in the selected customer folder

Procedure

1. **Save** the NextGen Controller IP address in the main window.
2. **Select** NGC EventLog Puller.
3. **Check** the Target IP shown in the module window.
4. **Select** Browse and choose an existing customer folder with write access.
5. **Select** Run EventPuller.
6. **Wait** until EventPuller completed successfully appears.
7. **Record** the full ZIP path shown in the result panel and transfer the archive according to the service procedure.

> **Archive handling:** The archive may contain customer and diagnostic data. Keep it protected and share it only through approved channels. Obtain the ZIP password through the authorized internal process.

9. Connection Requirements

- The service computer must have a working network connection to the target device.
- The correct Robot IP Address must be saved in the main interface.
- SSH/SFTP access to the controller must be available on TCP port 22 for modules that use it.
- The selected customer folder for an EventLog Puller must already exist and be writable.
- Firewall, VPN, or network-segmentation rules must permit the required connection.

10. Error Handling

Message or symptom	Check / action
Failed to connect	Verify the IP address, network path, controller power, and SSH availability.
Missing IP Address	Return to the main interface, enter the controller IP address, and select Save IP.
Missing Customer Folder	Select Browse and choose the folder in which the ZIP must be created.
Invalid Customer Folder	Choose an existing folder and confirm that the user has write access.
Event log folder or file was not found	Confirm that the correct LGC/NGC puller was selected and that the controller version provides the expected event logs.
Permission denied	Confirm controller access and use an authorized service account/build.
No event log files were found	Confirm the controller generation and verify that event logs exist.
Program is corrupted or lgc.dll/ngc.dll could not be decrypted	Repair or reinstall the application from an approved installer package.
Operation failed	Read the complete result panel, verify module compatibility, and confirm that required files and folders exist.

Escalation: If an error persists, preserve the exact result text and provide it to technical support. Do not repeatedly run a destructive module while troubleshooting.

11. Security Notes

- Controller credentials and protected configuration are handled internally by the application.
- External modification of application files, bundled DLL files, or embedded configuration is prohibited.
- EventLog Puller archives are password-protected but must still be treated as sensitive service data.
- Store and transmit archives only through approved customer and company channels.
- Delete local archive copies when retention is no longer required and company policy permits deletion.

12. Best Practices

- Verify the physical robot/controller and the saved IP address before every operation.
- Match the module color to the controller generation.
- Read the information panel and warnings before selecting Run or confirming an action.
- Perform backups whenever possible before permanent data removal, version manipulation, or recovery work.
- Use a clear customer-folder naming convention for EventLog Puller output.
- Do not close the application, disconnect the network, or restart the controller during an active operation unless instructed.
- Record the final result and output path in the service report.

13. Support

For technical support, approved installers, or software updates, contact:

Zwart Industrial Innovations

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